

# Strategy D is a dead end

the  $q$ -defect at  $V_F$  has wild ramification index  $p^{r-1}$ , a birational invariant; and why  
wildness *enables* absorption

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**Purpose.** Strategy D of `q_ramification_along_VF.tex` proposed to dodge the collision point by choosing a different resolution  $X'_{C'} \rightarrow C'^{(d)}$  in which  $\eta_{C'}$  is a codimension-one centre for  $q : (\mathcal{P}_{G/H})^{(d)} \rightarrow (\mathcal{P}_{G/H})^{[d]}$ , and then to apply the codimension-one unramifiedness lemma ?? iteratively. This note shows the idea cannot succeed, by a single invariant computation, and then settles the follow-up question of whether the (necessarily) wild  $q$ -defect can still absorb the ramification of  $E/F$ .

## Summary.

- **Theorem 1:**  $e(V_{C'} \mid V_F) = p^{r-1}$ . For  $r \geq 2$  this is wild ( $p \mid e$ ), so  $q$  is wildly ramified along  $V_F$ .
- **Corollary 2:** a ramification index of a fixed divisorial place is a birational invariant; no resolution changes  $p^{r-1}$  to 1. Strategy D is impossible, and the codimension-one lemma — which only ever produces  $e = 1$  — can never reach  $V_{C'}$ . This upgrades observation (O2) of the parent note from heuristic to theorem.
- **Section 4:** the wild index  $p^{r-1}$  does *not* block absorption; it is the mechanism that *permits* it. The real obstruction is not the index but a finer, leading-term coincidence, which is what the global hypotheses (B)+(C) must rule out.

## 1 The setup and the model

Notation as in `a_wrong_path.tex` and the parent note: after the standard reductions ( $k = \bar{k}$ ,  $\mathcal{P} \rightarrow U$  totally ramified at  $p_\infty$ , ??),  $G = \mathbb{Z}/p^r$ ,  $H \cong \mathbb{Z}/p$  the top layer,  $G/H \cong \mathbb{Z}/p^{r-1}$  the bottom layer,  $f : C' \rightarrow C$  the  $G/H$ -cover with  $p'_\infty \mapsto p_\infty$ . Write  $\varpi, u$  for uniformisers at  $p_\infty, p'_\infty$ ; total ramification of the bottom layer gives

$$\varpi = u^{p^{r-1}} \cdot (\text{unit}). \tag{1}$$

We use the *working* local model of the proven  $\mathbb{Z}/p$  case (`RamificationAfterBlowupIsTame.tex`, eq. ?? and the lines around it), *not* the local models of the post-mortems `a_wrong_path.tex` /

`inertia_symmetry_obstruction.tex` (those record dead ends and use mismatched valuations; e.g. the “ $J \supseteq G^d$ ” claim there would already contradict the proven  $\mathbb{Z}/p$  base case). In the working model the exceptional-divisor valuation of the blowup of a  $d$ -fold symmetric power at the collision point has an *explicit uniformiser*: the top elementary symmetric function. Thus

$$V_C : \quad \text{uniformiser } e_d = \prod_{i=1}^d \varpi_i, \quad v_C(e_d) = 1; \quad V_{C'} : \quad \text{uniformiser } \epsilon_d = \prod_{j=1}^d u_j, \quad v_{C'}(\epsilon_d) = 1, \quad (2)$$

where  $\varpi_i$  (resp.  $u_j$ ) are the base (resp. branch) coordinates of the  $d$  points. (Legitimate by the isomorphic-completion principle of `RamificationAfterBlowupIsTame.tex`: only  $\widehat{\mathcal{O}}_{C', p'_\infty} \cong k[[u]]$  enters.)

## 2 The ramification index of the $q$ -defect

**Theorem 1.**  $e(V_{C'} | V_F) = e(V_{C'} | V_C) = p^{r-1}$ . In particular, for  $r \geq 2$  the extension  $K_{C'}/F$  is wildly ramified at  $V_{C'} | V_F$ .

*Proof.* By (1),  $\varpi_i = u_i^{p^{r-1}} \cdot (\text{unit})$  for each of the  $d$  branches, so the base uniformiser of  $V_C$  pulls back to

$$e_d = \prod_{i=1}^d \varpi_i = \left( \prod_{j=1}^d u_j \right)^{p^{r-1}} \cdot (\text{unit}) = \epsilon_d^{p^{r-1}} \cdot (\text{unit}).$$

Hence  $v_{C'}(e_d) = p^{r-1} v_{C'}(\epsilon_d) = p^{r-1}$  by (2). Since  $e_d$  is a uniformiser of  $V_C$  and  $\eta_{C'} \mapsto \eta_C$  under  $f^{(d)}$  (so  $V_{C'} | V_C$ ), the ramification index is

$$e(V_{C'} | V_C) = v_{C'}(\text{uniformiser of } V_C) = v_{C'}(e_d) = p^{r-1}.$$

Finally  $F/K_C$  is unramified at  $V_C$  — this is hypothesis (H1), equivalently the  $\mathbb{Z}/p^{r-1}$ -symmetrised datum  $\sum_i b(\varpi_i)$  lies in the residue field  $\kappa_C$  by the bound  $m < d$  (the lemma of the working  $\mathbb{Z}/p$  proof) — so  $e(V_F | V_C) = 1$  and  $e(V_{C'} | V_F) = e(V_{C'} | V_C)/e(V_F | V_C) = p^{r-1}$ .  $\square$

*Remark.* The same computation explains why ?? succeeds at boundary *divisors* and only there. At the generic point of a boundary divisor  $Z_x$  exactly *one* of the  $d$  points sits at  $p'_\infty$  and the other  $d-1$  are free; only one factor  $u_j$  contributes to the collision, the symmetric structure retains  $d-1$  free copies of  $K(C')$  in the residue field, and one finds  $e = 1$  (trivial inertia). At  $V_{C'}$  all  $d$  points collide through the degree- $p^{r-1}$  wild cover, and (1) forces  $e = p^{r-1}$ . The codimension-one lemma is therefore not “too weak to reach  $V_{C'}$ ”; its conclusion is simply *false* there.

## 3 Why Strategy D cannot be repaired

**Corollary 2.** *There is no birational model on which  $q$  is unramified at the place  $V_{C'} | V_F$ , and no factorisation of  $q$  into codimension-one steps to which ?? applies and whose composite recovers  $V_{C'}$ . Strategy D is impossible.*

*Proof.* The ramification index  $e(w | w_F)$  of a fixed place  $w = V_{C'}$  of  $K_{C'}$  over its restriction  $w_F = V_F$  to  $F$  is an invariant of the *valuations*  $w, w_F$  alone: it equals  $[|w(K_{C'}^\times)| : |w_F(F^\times)|]$ , the index of value groups, and does not refer to any model. (Equivalently:  $w$  and  $w_F$  are intrinsic points of

the Riemann–Zariski spaces of  $K_{C'}$  and  $F$ ; a resolution only re-presents  $w$  as a divisor, it cannot alter  $w$ .) By [Theorem 1](#) this index is  $p^{r-1} > 1$ . Any application of ?? yields a place with  $e = 1$ ; a composite of such steps landing on  $V_{C'}$  would give  $e(V_{C'} \mid V_F) = 1$ , contradicting [Theorem 1](#).  $\square$

So the only thing a resolution can buy is to *realise*  $V_{C'}$  as a nice divisor for *computing* its higher ramification (a service to Strategy A), never to make  $q$  unramified there. Strategy D, as a route to the theorem, is closed.

## 4 Does a wild $q$ -defect still absorb $E/F$ ? Yes — that is the point

It is tempting to hope that a *large* wild defect  $e = p^{r-1}$  is “too violent” to let (H2) coexist with a ramified  $E/F$ , i.e. that wildness blocks absorption. The opposite is true: wildness is the *enabling* mechanism.

**The mechanism.** Present the top layer locally as Artin–Schreier over  $\widehat{F}$ ,

$$\widehat{E} = \widehat{F}(y), \quad y^p - y = a, \quad v_F(a) = -\beta_E, \quad \gcd(\beta_E, p) = 1,$$

so  $E/F$  is (if ramified) totally wildly ramified of break  $\beta_E \geq 1$  (??(3)). Base-change along  $\widehat{F} \hookrightarrow \widehat{K}_{C'}$ , of ramification index  $e = p^{r-1}$  ([Theorem 1](#)). The *same* element  $a$  now has

$$v_{C'}(a) = e \cdot v_F(a) = -p^{r-1}\beta_E, \tag{3}$$

and the pole order  $p^{r-1}\beta_E$  is **divisible by  $p$**  (as  $r \geq 2$ ). Divisibility by  $p$  is exactly the trigger for Artin–Schreier reduction over  $\widehat{K}_{C'}$ : writing  $\pi$  for a uniformiser of  $\widehat{K}_{C'}$  and  $a = c\pi^{-p^{r-1}\beta_E} + \dots$  with  $c = c_0^p$  ( $k$  algebraically closed), the substitution  $y \mapsto y - c_0\pi^{-p^{r-2}\beta_E}$  replaces  $a$  by

$$a - \wp(c_0\pi^{-p^{r-2}\beta_E}) = a - c_0^p\pi^{-p^{r-1}\beta_E} + c_0\pi^{-p^{r-2}\beta_E},$$

killing the leading pole. Iterating, the pole order can descend  $p^{r-1}\beta_E \rightarrow p^{r-2}\beta_E \rightarrow \dots$ , and *if the lower-order data cooperate it can reach  $\leq 0$* , i.e.  $\widehat{E}\widehat{K}_{C'}/\widehat{K}_{C'}$  becomes unramified — (H2) holds while  $E/F$  is ramified. This is precisely the coincidence  $\widehat{K}_{C'} = \widehat{E}$  engineered in ??. Over  $\widehat{F}$  no such reduction is available, because there  $v_F(a) = -\beta_E$  is *prime to  $p$*  and  $\widehat{F}$  has no elements of the intermediate  $\pi$ -valuations  $-p^j\beta_E$  ( $j < r-1$ ) used by the substitutions; the wild base change is exactly what supplies them.

*The index  $p^{r-1}$  neither causes nor prevents absorption by itself; it is the gate that makes absorption arithmetically possible. Absorption completes only when the reduction terminates at a non-positive pole, which is a leading-coefficient condition, equivalent to  $\widehat{E} \subseteq \widehat{K}_{C'}$ .*

### Consequences for the live strategies.

1. The correct target is *not* “ $\beta_q < \beta_E$ ” as a bare inequality of indices, but the finer statement that the reduction of  $a$  over  $\widehat{K}_{C'}$  **halts with a positive break**, equivalently  $\widehat{E} \not\subseteq \widehat{K}_{C'}$ . Generically the reduction halts at break  $\beta_E$  (the prime-to- $p$  tail  $c_0\pi^{-\beta_E}$  cannot be removed), so  $E \cdot K_{C'}/K_{C'}$  *stays ramified* and (H2) forces the Goal; absorption is the non-generic, conspiratorial case.

2. This is exactly Strategy C of the parent note, now with a precise criterion: show the global hypotheses

$$(B) \text{ ram}(\mathcal{P}) \leq d, \quad (C) \ r < d$$

force the top-layer Artin–Schreier class  $[a] \in \widehat{F}/\wp(\widehat{F})$  to remain nontrivial after the base-change  $\widehat{F} \hookrightarrow \widehat{K_{C'}}$ , i.e. to survive in  $\widehat{K_{C'}}/\wp(\widehat{K_{C'}})$ . The class  $[a]$  is built from the *top* Witt component  $a_{r-1}$  (break  $\beta_E \sim u_r$ ), while  $\widehat{K_{C'}}$  is built from the *bottom* cover (breaks  $u_1, \dots, u_{r-1}$ ); by Hasse–Arf  $u_{r-1} < u_r$ , and (B) keeps all  $u_i < d$  while (C) keeps the number of layers below the blow-up scale. The task is to convert this separation of sources into non-containment of the local extensions.

3. Strategy A is unaffected and remains the way to *produce* the number: compute the reduced break of  $a$  over  $\widehat{K_{C'}}$  directly in the symmetric-power model. [Theorem 1](#) gives its starting input ( $v_{C'}(a) = -p^{r-1}\beta_E$ ) for free.

## 5 Takeaway

- **Strategy D is dead**, by the invariant  $e(V_{C'} \mid V_F) = p^{r-1}$  ([Theorem 1](#) and [corollary 2](#)). A fixed place cannot be de-ramified by re-resolving.
- **Wildness enables, not blocks, absorption** ([Section 4](#)): the base change multiplies the break to  $p^{r-1}\beta_E \equiv 0 \pmod{p}$ , opening Artin–Schreier reduction. The genuine obstruction lives one order deeper, in the leading coefficients — i.e. in whether  $\widehat{E} \subseteq \widehat{K_{C'}}$ .
- **The fight is over a class in  $\widehat{F}/\wp(\widehat{F})$** , not over a ramification index. (B)+(C) must keep the top-layer class nontrivial over  $\widehat{K_{C'}}$ ; the small decisive computation of the parent note ( $r = 2$ ,  $d = 3$ ) is precisely the test of this.